

Proposed Solution

Date	29 June 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report details our AI-powered web dashboard that maps event descriptions into conversational icebreakers using modular NLP pipelines.

Project Overview

Field	Details
Objective	Provide users with an instant tool to generate creative, personalized networking conversation starters based on event contexts and personal goals.
Scope	Zero-shot theme extraction from event text, sentence completion using GPT-2 small, Wikipedia fact validation, and local session auditing.
Problem Description	Networking is essential but many struggle with social anxiety and face difficulties identifying overlap in technical topics during conferences.
Problem Impact	Missed career connections, wasted registration fees, and poor team building experiences.
Approach	Utilize a FastAPI REST backend wrapping DistilBERT classifier and GPT-2 pipeline, feeding a Streamlit interactive dashboard UI.
Key Features	• Theme Extraction • Conversation Prompt Generator • Wikipedia Fact Check • Session History & Feedback Loggers

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware: Computing	CPU/GPU requirements	Intel i3/i5 or higher, GPU not required.
Hardware: Memory/Storage	RAM & Disk specifications	Minimum 4GB RAM (8GB recommended), 10GB free disk space.
Software: Frameworks	Web backend & frontend	FastAPI (Backend), Streamlit (Frontend).
Software: Libraries	Core NLP and API libraries	transformers (HuggingFace), torch, requests, pytest.
Software: Environment	IDE choice	Visual Studio Code, PyCharm.

Data	Pre-trained model source	DistilBERT zero-shot MNLI and GPT-2 weights (HuggingFace hub).
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Solution Scope and NLP Architecture Description:

The system is engineered as a fully local full-stack AI deployment. By hosting the machine learning models (DistilBERT and GPT-2) inside the local python process instead of calling commercial paid APIs (like OpenAI GPT-4), we guarantee complete data privacy, offline usability, and eliminate ongoing operational API token costs. This approach makes the Personalized Networking Assistant ideal for deployment in highly active settings (like large conferences) where network connectivity is spotty. The modular backend structure separates routers, services, and models, facilitating simple future expansion to larger local architectures or custom-trained enterprise weights.